

Florida Nonnative Reptiles

Distribution, Establishment, Husbandry, and Management

GIS, Power BI, and Field Case Study Portfolio Project

Abstract

Florida supports an unusually large number of established nonnative reptiles. Their distributions reflect decades of escaped and released pets, repeated introductions, suitable climate, urban development, and successful reproduction. This project examines reported occurrences of five species in Florida from 2010 through 2025: Burmese python (*Python bivittatus*), green iguana (*Iguana iguana*), Argentine black and white tegu (*Salvator merianae*), Peter's rock agama (*Agama picticauda*), and Nile monitor (*Varanus niloticus*).

A total of 55,033 EDDMapS records were assembled and cleaned for analysis. Of these, 51,760 contained usable coordinates for mapping in QGIS. Power BI was used to compare species totals, county totals, and temporal patterns. The resulting maps show a strong concentration of records in South Florida, along with additional clusters on the Atlantic coast, in southwest Florida, and around Tampa Bay and Central Florida.

The project grew from my experience raising an Argentine black and white tegu named Ghost, an animal I acquired shortly before Florida prohibited new personal ownership of tegus. Caring for a large reptile over several years made the connection between exotic animal ownership and invasive species management difficult to ignore. A second case study developed during the project when I began observing Peter's rock agamas at an urban site in Tampa, attempted live capture, and successfully removed a small juvenile.

The statewide data and the two case studies examine the same issue at different scales. The maps document where nonnative reptiles have been reported and where concentrations have developed. The case studies examine how those populations can begin, what management looks like in the field, and what responsibility follows when an individual animal remains alive in human care.

Introduction

My interest in Florida's invasive reptiles began with a pet.

I acquired Ghost, an Argentine black and white tegu, as a juvenile shortly before Florida restricted personal ownership of the species. He is now approximately four years old and remains legally maintained under Florida's grandfathered permitting system.

As he grew, I became increasingly interested in why tegus had become such a serious management concern in Florida. Ghost was an individual animal I knew well, but wild animals of the same species were being trapped and removed because established populations threatened native wildlife.

The practical side of keeping him also made the problem easier to understand. An adult tegu bears little resemblance to the relatively manageable juvenile somebody might originally purchase. Ghost eventually required a custom eight-foot wooden enclosure, substantial heating and UVB equipment, a varied omnivorous diet, and veterinary treatment that has included multiple surgeries.

It was easy to imagine how an owner who was poorly prepared for that commitment could eventually want to get rid of the animal.

Florida wildlife managers believe the state's established tegu populations originated from escaped or intentionally released captive animals. Reproducing populations are now established in Hillsborough, Miami-Dade, and Charlotte counties, with an emerging population documented in St. Lucie County. FWC considers the Argentine black and white tegu invasive because of its effects on native wildlife (Florida Fish and Wildlife Conservation Commission [FWC], 2026a).

That history led me to broader questions. Where are Florida's nonnative reptile populations concentrated? How do the distributions of different species compare? Which counties receive the most reports? Are

reported ranges changing over time? Can occurrence databases help distinguish an isolated animal from an area where a population has become established?

I chose five species for the project: Burmese python, green iguana, Argentine black and white tegu, Peter's rock agama, and Nile monitor.

The project later gained a field component when I found a population of Peter's rock agamas at a commercial property in Tampa. Instead of seeing the species as coordinates in an occurrence database, I could watch several animals occupying the same small area, using drainage infrastructure, pavement, trees, and other pieces of an ordinary urban landscape.

I eventually attempted live capture at the site and successfully removed a very small juvenile.

That experience changed how I thought about the data. A point on a map is useful because it records a species at a location. At the site itself, that point may belong to a population containing multiple animals, different age classes, preferred refuges, basking areas, and patterns of activity that the coordinate alone cannot describe.

Data and Methods

Occurrence records were obtained from EDDMapS, the Early Detection and Distribution Mapping System, which collects and maps reports of invasive and nonnative species from agencies, researchers, organizations, and members of the public. EDDMapS records available for mapping and download are reviewed by verifiers or entered as expert data (EDDMapS, 2026).

The query covered Florida from January 1, 2010 through December 31, 2025. Original records with Positive or Treated infestation status were retained.

The combined dataset contained 55,033 records:

Species	Records	Share of dataset
Burmese python	19,240	35.0%
Green iguana	12,870	23.4%
Argentine black and white tegu	11,955	21.7%
Peter's rock agama	9,678	17.6%
Nile monitor	1,290	2.3%
Total	55,033	100%

The master dataset preserved records even when coordinates were unavailable so that they could still contribute to non-spatial summaries. A separate QGIS-ready dataset contained the 51,760 records with usable latitude and longitude, or about 94.1 percent of the original records.

Coordinate coverage varied by species. Burmese python, green iguana, tegu, and Nile monitor records all had coordinate coverage above 95 percent. Peter's rock agama was an exception. Only 7,587 of its 9,678 records contained usable coordinates, a coverage rate of approximately 78.4 percent. Agama records therefore make up a smaller share of the mapped dataset than of the full database.

The cleaned point data were loaded into QGIS and compared with Florida county boundaries. County totals were calculated spatially so that mapped coordinates, rather than only the county names supplied in source records, determined county membership.

Data quality checks also produced several useful examples of the problems that can appear in large occurrence datasets. One tegu record contained a positive longitude that placed the animal outside its reported Florida location. Changing the sign from +80.50196 to -80.50196 placed the point in Miami-Dade

County and matched the accompanying location information. The correction was documented rather than silently altering the source record.

Eleven otherwise usable points fell outside Florida county polygons and remained visible as offshore anomalies. Seven belonged to green iguanas, two to Burmese pythons, one to Peter's rock agama, and one to a Nile monitor. Some were only slightly offshore, while two python records appeared much farther from land. These records were retained as examples of spatial uncertainty rather than deleted simply because they looked inconvenient.

QGIS was used to create a statewide species map and a county-level choropleth based on the number of mapped records falling inside each county. The project also included a case-study layer identifying the agama field site, the corrected tegu coordinate, and the offshore anomalies.

The same cleaned data were imported into Power BI to provide interactive species, county, and year filters and to compare report totals across the dataset.

Results

Species composition

Burmese pythons were the most frequently represented species in the complete dataset, with 19,240 records, or 35 percent of all observations.

Green iguanas were second with 12,870 records, followed by Argentine black and white tegus with 11,955. Peter's rock agamas accounted for 9,678 records, while Nile monitors were much less common in the dataset at 1,290.

The total for Peter's rock agama deserves some context. Recent Florida research describes the agama as the most frequently reported nonnative reptile to FWC since 2021. My dataset covers a longer period beginning in 2010 and includes records originating from multiple reporting channels and bulk datasets. The difference is a reminder that rankings depend on the database, time period, and reporting system being analyzed (Claunch et al., 2026).

[Insert Figure 1 here]

Figure 1. Power BI species-count visualization showing total records for the five selected reptile species.

Geographic concentration

The county analysis showed a pronounced geographic concentration.

Of the 51,760 records with usable coordinates, 51,749 fell inside Florida county boundaries. Miami-Dade contained 22,247 of those points, approximately 43 percent of all county-assigned mapped observations.

The highest county totals were:

County	Mapped records
Miami-Dade	22,247
Collier	6,698
Broward	4,898
Palm Beach	4,520
Monroe	2,574
St. Lucie	2,455
Lee	1,728
Martin	1,201
Charlotte	1,127

Brevard	650
Hillsborough	528

These eleven counties accounted for roughly 94 percent of all county-assigned mapped records.

The overall map is therefore heavily weighted toward southern Florida. Miami-Dade is dominant, but the distribution is not limited to a single region. The species map also shows concentrations along the Atlantic coast, in southwest Florida, and around parts of Central Florida and Tampa Bay.

[Insert Figure 2 here]

Figure 2. QGIS statewide occurrence map showing individual records symbolized by species.

[Insert Figure 3 here]

Figure 3. QGIS county choropleth showing total mapped records by Florida county.

The regional pattern changes when individual species are filtered.

Burmese python records are heavily associated with South Florida and the Everglades region. Green iguana records are also strongly concentrated in southern Florida, particularly developed and coastal areas. Nile monitor records form a much smaller dataset and are associated with several known southwest and South Florida populations.

Tegus are useful for showing how several separate population centers can appear within the same state. FWC recognizes established reproducing populations in Hillsborough, Miami-Dade, and Charlotte counties and an emerging population in St. Lucie County (FWC, 2026a).

Peter's rock agama has a broader patchwork distribution. FWC states that it has become established in multiple counties across much of southern and central Florida, while UF/IFAS describes numerous populations occupying developed environments (FWC, 2026b; Gioeli & Johnson, 2025).

These species-specific differences matter more than the statewide county ranking by itself. Miami-Dade's enormous total tells us that South Florida dominates the combined dataset, but filtering by species reveals different invasion histories within that broader pattern.

Reports through time

The dataset includes observation dates from 2010 through 2025, allowing the Power BI report to be filtered by year and species.

Annual record counts are useful for examining changes in reporting, but I have avoided treating them as direct population estimates. EDDMapS combines observations from several source types, including public reports and bulk records from government agencies. Reporting intensity can change even when biological abundance does not.

A rise in records may result from population expansion, increased public awareness, additional agency monitoring, changes in reporting technology, or several of these at once.

Spatial change is therefore especially important when interpreting the year filter. When later observations begin appearing repeatedly outside an older geographic concentration, they provide stronger evidence of possible range expansion than a simple increase in the number of reports from an existing population center.

The interactive dashboard allows those patterns to be explored rather than reducing the entire sixteen-year dataset to a single statewide growth rate.

[Insert Figure 4 here]

Figure 4. Power BI annual-report trend with species, county, and year filters.

Data quality and reporting bias

The mapping process also exposed weaknesses in occurrence data that are easy to overlook when working only from a spreadsheet.

Approximately 5.9 percent of the master records could not be mapped because coordinates were missing. The problem was particularly important for Peter's rock agama, which accounted for 2,091 of the 3,273 records without usable coordinates.

Spatial errors were present even among records that did contain coordinates. The corrected tegu longitude and eleven offshore points were obvious examples because the coordinates could be checked against county boundaries and Florida's coastline.

Less obvious errors may remain.

Reporting effort is another source of bias. A conspicuous lizard living beside a parking lot is much easier to encounter than an animal occupying remote wetlands. Counties with large human populations also contain more potential observers. Smartphones and reporting applications have made georeferenced observations increasingly easy to submit. EDDMapS itself allows users to report directly from mobile devices using GPS (EDDMapS, 2026).

The resulting maps are maps of reported occurrence. They provide useful evidence about distribution without functioning as a census.

Case Study 1: Ghost and the Origin of the Project

Ghost was my introduction to the subject before I ever began downloading occurrence data.

I acquired him as a juvenile shortly before tegus were added to Florida's Prohibited Species list in April 2021. FWC allowed qualifying animals already legally possessed at the time of the rule change to remain with their owners under grandfathered personal-use permits (FWC, 2026a).

I have now raised Ghost for approximately four years.

He is an albino Argentine black and white tegu and lives in a custom-built wooden enclosure approximately eight feet long. His care includes controlled heat and UVB, humidity, secure housing, and an omnivorous diet that has included frozen-thawed rodents, cooked eggs, and vegetables.

The scale of his care changed enormously as he grew.

That experience helped me understand one of the pathways behind Florida's nonnative reptile populations. A juvenile exotic reptile can be relatively inexpensive and physically manageable. The same animal several years later may require a room-sized enclosure, large quantities of food, specialized equipment, and access to a veterinarian comfortable treating large reptiles.

FWC reports that Argentine black and white tegus can approach five feet in length and may live for as long as twenty years. Females can lay an average of about 35 eggs annually. Florida's reproducing populations are believed to have originated through escapes or deliberate releases of captive animals (FWC, 2026a).

Ghost has also required significant veterinary care. He developed pigment-cell tumors and has undergone multiple surgeries to remove them. Reptile chromatophoromas are documented in veterinary literature, but their causes are often uncertain. His albinism and history of artificial UVB exposure raise questions that cannot be resolved from one animal, so I do not attribute the tumors to either factor (Monahan et al., 2022).

[Insert Figure 5 here]

Figure 5. Ghost, an Argentine black and white tegu maintained under Florida's grandfathered permitting system.

[Insert Figure 6 here]

Figure 6. Ghost's custom eight-foot enclosure, showing the scale of long-term adult tegu husbandry.

The burden of keeping Ghost has never made me consider releasing him, but it has made the behavior behind some releases easier to understand.

A person who buys a small tegu without appreciating its adult requirements may eventually face an animal they cannot house or afford. Releasing that animal transfers the owner's problem to the surrounding ecosystem.

Florida's Exotic Pet Amnesty Program exists as an alternative. The program accepts nonnative pets year-round, including Conditional and Prohibited species, and connects owners with qualified adopters (FWC, 2026c).

Ghost also changed the way I react to wild tegu control.

An acquaintance at my local parish once mentioned that a tegu had appeared on his property and that he shot it.

Florida allows landowners to humanely kill wild tegus on private property, and FWC actively encourages their removal because of the risk established populations pose to native wildlife. Thousands have been removed through FWC and partner programs (FWC, 2026a).

I understood why he had done it and still found the story discouraging.

After living with a tegu for years, it is difficult for me to see one as interchangeable with a dot in an invasive species dataset. Tegus are behaviorally complex reptiles. Controlled studies have shown that Argentine black and white tegus can follow conspecific chemical trails and use those cues when making navigation decisions (Parker et al., 2023; Richard et al., 2020).

The wild animal on that property did not understand that it belonged to a prohibited species. It was behaving normally in an environment where humans had introduced its species.

That does not produce an easy alternative.

Capturing every wild tegu and maintaining it permanently would be impractical. My experience with one adult makes the scale of such a proposal obvious. Thousands of large reptiles would require secure enclosures, food, staff, veterinary care, land, and long-term funding. Moving them to another uncontrolled location would simply move the ecological risk.

Population control therefore has to remain practical even when the individual animals are worth caring about.

That tension became one of the central ideas of this project. Prevention is preferable partly because establishment eventually creates situations where otherwise healthy animals must be removed or euthanized to protect an ecosystem.

Case Study 2: Peter's Rock Agama Field Study

The Peter's rock agama case began much closer to the type of data represented in the GIS project.

I had repeatedly seen agamas around a commercial property in Tampa. After several visits, it became clear that the site supported more than an occasional individual. I sometimes observed at least five animals active at once, including different size classes.

The property was heavily developed. Agamas used pavement, trees, curbs, and especially a drainage area containing numerous hiding places.

This habitat matched published descriptions of Florida Peter's rock agama populations. UF/IFAS notes that the species frequently occupies urban and suburban environments and uses buildings, sidewalks, curbs, fences, trees, and similar structures (Gioeli & Johnson, 2025).

Seeing the animals in person made detection feel much less absolute than it appears in a GIS layer.

Some visits produced several visible animals. Rain could cause them to disappear. Their speed made close approaches difficult. Activity changed with weather and time of day.

A population can therefore be present even when an observer fails to detect it during a particular visit.

Capture attempts

I received permission from the business to place traps on the property.

My primary method used collapsible fabric funnel traps baited with live crickets. One was positioned near an area where agamas had been observed around a tree, while another was placed near the drainage structure where activity was particularly common.

The approach closely resembled one evaluated in a 2026 Florida study by Claunch and colleagues. Their experiment compared fabric minnow traps, lizard-fishing, and air-rifle removal. The traps were baited with crickets and deployed for one-hour periods. Six agamas were captured by traps during the study, although trapping was less efficient than air-rifle removal. The researchers also observed agamas approaching or sitting on traps without successfully finding the funnel openings (Claunch et al., 2026).

My own attempts quickly produced several practical lessons.

Rain interrupted one session and caused visible agama activity to decline. The published removal study likewise avoided rain because the animals retreated and became unavailable for removal during rain events (Claunch et al., 2026).

I also left traps at the site overnight once. One was stolen. The other remained in the drainage area but was empty the next morning and contained numerous partially eaten cricket remains.

The result was impossible to interpret confidently. An agama may have entered, fed, and eventually found its way out. Another animal may have eaten the crickets. The long unattended period removed the ability to distinguish among those possibilities.

For future work, I would use shorter supervised trapping periods while agamas are already visibly active.

A small juvenile, approximately the size of a small anole, was eventually captured during the field effort and transferred to captivity.

[Insert Figure 7 here]

Figure 7. Urban Peter's rock agama observation site in Tampa, including the drainage and basking habitat.

[Insert Figure 8 here]

Figure 8. Cricket-baited collapsible funnel trap used during field capture attempts.

[Insert Figure 9 here]

Figure 9. Captured juvenile Peter's rock agama shortly after removal from the field site.

Early captive acclimation

The juvenile entered a temporary 10-gallon quarantine enclosure because an adult-sized enclosure would have made close observation of such a small animal unnecessarily difficult.

Early husbandry involved establishing a suitable thermal gradient, UVB lighting, secure housing, and reliable feeding.

An existing 100-watt intense basking bulb proved too powerful for prolonged use over the small aquarium. The basking surface could reach approximately 100 degrees Fahrenheit in around ten minutes. That required careful monitoring until lower-output equipment could be used.

The enclosure initially received daylight through a nearby window, but visible sunlight through glass could not substitute for a proper UVB source.

Feeding required its own adjustment.

I already maintained colonies of discoid roaches and expected small nymphs to make an easy feeder. The juvenile repeatedly ignored them, even when they were placed in a smooth-sided feeding dish where they could not immediately bury themselves.

Waxworms were accepted first. Mealworms followed later, allowing calcium supplementation to be introduced on prey the animal already recognized.

The details are ordinary husbandry problems, but experiencing them after a field capture gave me another perspective on live removal.

Before capture, the juvenile was one small member of a nonnative population.

After capture, keeping it alive meant meeting the requirements of that individual animal. Food that was theoretically appropriate did little good if it refused to eat it. A heat lamp marketed for reptiles still had to be adjusted to the size of the enclosure. Human contact had to be introduced gradually because the animal remained highly wary.

The transition from wild animal to captive animal was therefore an acclimation process rather than simply a change of location.

[Insert Figure 10 here]

Figure 10. Temporary juvenile agama enclosure showing shelter, substrate, basking area, and feeder dish.

From Introduction to Establishment

Ghost and the agama represent opposite directions through the same human-created system.

Ghost entered Florida through the exotic pet trade and has remained in captivity.

Florida's wild tegus descended from animals that did not.

The juvenile agama was born into a Florida population descended from animals introduced through the pet trade decades earlier and was then removed from that population into captivity. FWC traces the state's Peter's rock agama populations to escaped or released pets beginning in the 1970s (FWC, 2026b).

The GIS data capture the geographic consequences after those introductions occur.

An escaped reptile may die without reproducing. Another may survive but remain isolated. If multiple animals encounter one another in suitable habitat and reproduce successfully, their descendants can create a local cluster of observations.

Continued reproduction and dispersal can gradually turn that local cluster into a regional distribution.

The maps suggest different stages and histories among the five species. Some have dense South Florida concentrations. Tegus have several geographically separated population centers. Peter's rock agamas appear across a broad part of the peninsula. Nile monitors remain much less frequently reported in this dataset.

Climate matters, but human geography matters too.

Urban and suburban areas provide heat, structures, food, disturbed habitat, and constant opportunities for people to move animals. They also provide observers, which complicates interpretation because the same development that may favor an introduced reptile can also increase the chance that somebody reports it.

Environmental Impact and Management

Nonnative status alone does not tell us how damaging a species is.

Peter's rock agama is clearly established in Florida, but UF/IFAS notes that evidence of substantial ecological or economic harm remains limited. FWC nevertheless identifies potential concerns involving predation and competition, especially where populations become dense (Gioeli & Johnson, 2025; FWC, 2026b).

Argentine black and white tegus have a stronger record of documented impact.

FWC reports that tegus consume eggs and small animals and have been documented eating American alligator eggs and threatened juvenile gopher tortoises. Their broad diet, high reproductive output, and ability to survive in disturbed habitats make established populations difficult to ignore (FWC, 2026a).

Management therefore has to consider the species, location, size of the population, and strength of the evidence for harm.

For a new introduction, rapid detection and removal may prevent establishment.

For a localized breeding population, sustained trapping or targeted removal may still reduce numbers or contain expansion.

Once a species becomes widespread, statewide eradication may no longer be realistic. Management may instead focus on local suppression, protecting sensitive habitats, preventing additional introductions, and monitoring newly occupied areas.

The 2026 agama removal study provides a useful example. Thirty-seven agamas were removed during 36 hours of experimental effort using three methods. Air rifles accounted for 27, traps for six, and lizard-fishing for four. The authors found all methods potentially useful but concluded that considerably more work would be needed to evaluate population reduction at larger scales (Claunch et al., 2026).

The practical difficulty becomes obvious once a species is numerous.

That is why early detection matters. FWC specifically asks observers to provide a photograph, location, and date when reporting nonnative wildlife because credible reports can help managers respond to new occurrences (FWC, 2026d).

Prevention remains even earlier in the process.

Secure captive housing reduces escapes. Better education about adult size, lifespan, and husbandry requirements may reduce deliberate releases. Rehoming programs give owners an alternative when they can no longer maintain an exotic animal (FWC, 2026c).

These measures are mundane compared with removing thousands of reptiles from the landscape, which is exactly why they are preferable.

Individual Welfare and Population Control

Working with occurrence data makes it easy to think at the population level. That is necessary when the question involves tens of thousands of records.

My experiences with the animals made the same subject more concrete.

Ghost is a member of a species that Florida is actively trying to remove from the wild, but my experience of him is an individual one. I know how he behaves, what he eats, how much space he requires, and what it means when he needs surgery.

That background made hearing about a wild tegu being shot uncomfortable even though I understood the management reasoning.

The agama produced a similar shift from the other direction. At the field site, it was one animal among several. After I removed the juvenile alive, its ecological classification did not tell me how to feed it or how much heat it needed.

There is no practical management program in which every invasive reptile can be relocated into permanent captive care.

There is also no ecological reason to treat avoidable suffering as useful.

Humane euthanasia, careful trapping practices, secure ownership, surrender programs, and prevention of future releases allow population management to remain practical without pretending the animals involved have no individual value.

The reptiles did not decide to become invasive. They are behaving according to their biology in environments where people placed them.

That does not remove the environmental consequences.

It does put the responsibility for dealing with those consequences largely on us.

Limitations

This project uses occurrence records rather than standardized population surveys.

The 55,033 EDDMapS records provide substantial geographic information, but observations are influenced by reporting effort, human population density, accessibility, species visibility, agency sampling, and changes in reporting technology.

The dataset also contains unequal spatial completeness among species. Peter's rock agama had substantially more missing coordinates than the other four species, which limits direct comparisons based only on mapped records.

County totals should therefore be interpreted as concentrations of reported observations rather than estimates of true animal abundance.

Temporal patterns require similar caution. An increase in annual reports can occur because a population is expanding, because more people are reporting it, because agencies increase survey effort, or because records from another source are added to the database.

The field case study is also limited.

I did not begin observing the Tampa agama site under a standardized survey design, and I did not record every early sighting. The observations cannot estimate the population size at the site.

The captured juvenile is one animal. Its feeding behavior and acclimation should not be generalized into species-wide husbandry conclusions.

Ghost is also one individual, and his medical history cannot describe the experience of every captive tegu.

The case studies are included because they provide biological context for the GIS analysis and help explain introduction and management pathways that are difficult to represent in a map.

Conclusion

When I began working with invasive reptile data, most of my questions were geographic. I wanted to know where the animals were being reported, which species were most common in the dataset, and whether recognizable populations appeared in different parts of Florida.

The completed dataset produced some clear patterns.

Burmese pythons had the largest number of records among the five selected species. Miami-Dade contained by far the greatest concentration of mapped observations, and the eleven highest-count counties accounted for roughly 94 percent of all points assigned to Florida counties. South Florida dominates the combined map, while species filtering reveals additional Atlantic coast, southwest Florida, Tampa Bay, and Central Florida populations.

The analysis also exposed the problems that accompany real occurrence data. Thousands of records lacked coordinates. Several points fell offshore. One tegu record contained a longitude sign error that had to be investigated rather than automatically trusted. Reporting effort varies between species, counties, and years.

Those imperfections do not make the data useless. They define what questions the data can answer responsibly.

The QGIS maps provide a statewide view of reported distribution. Power BI makes it possible to compare species, counties, and years interactively. The case studies supplied something different.

Raising Ghost was what first made me curious about the statistics behind Florida's invasive reptiles. His species went from exotic pet to prohibited invasive wildlife within the context of my own ownership, and caring for him showed how demanding large reptile husbandry can become.

The agama field study brought the project outside. I could visit an actual population represented by the type of records I had been mapping, observe the habitat the animals were using, attempt capture, and eventually remove one juvenile alive.

The larger management problem is difficult partly because the two scales cannot be separated completely.

Florida has populations that may damage native ecosystems and require control. Those populations are made of individual animals that did not choose to be introduced.

Once establishment has occurred, practical wildlife management may require trapping and killing animals because permanent captive placement is not realistic at population scale. The better outcome is to prevent that point from being reached whenever possible.

Occurrence reporting, GIS analysis, early removal, secure exotic-animal ownership, realistic husbandry education, and surrender programs all address different parts of that process.

The maps tell us where introductions have left a geographic footprint. Following that footprint back to the animals makes the reasons for preventing the next one much harder to ignore.

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Figure and Visual Plan

Figure	Visual	Priority	What it should show
Figure 1	Power BI species-count visualization	Required	Use the final dashboard visual comparing the five species totals.
Figure 2	QGIS statewide occurrence map by species	Required	Main spatial figure. Show all five species with a readable legend.
Figure 3	QGIS county choropleth	Required	Show total mapped records by county; Miami-Dade should stand out clearly.
Figure 4	Power BI annual trend	Required	Show records by year with species/county filters visible if possible.
Figure 5	Ghost portrait	Recommended	Clear photograph of Ghost showing the adult animal.
Figure 6	Ghost's eight-foot enclosure	Recommended	Wide photo that communicates the scale of adult tegu husbandry.
Figure 7	Agama field site	Recommended	Photo of the drainage/urban habitat; avoid prominently displaying business branding if possible.
Figure 8	Cricket-baited funnel trap	Recommended	Photo showing the trap setup used during field work.
Figure 9	Captured juvenile Peter's rock agama	Required for case study	Clear specimen photo with enough context to show its small size.
Figure 10	Juvenile quarantine enclosure	Optional	Useful if visually clean; shows how the field capture became a husbandry case.